Horizon Capital: A Multi-Agent Autonomous Trading System

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Abstract—Claims about LLM-driven trading are cheap to make and expensive to audit. Published backtests inherit survivorship bias and selection on the data used to report them, and the language model's contribution is seldom separated from the mechanical consequence of trading at a different rate. This paper reports the audit. Horizon Capital is a running autonomous trading system laid out as an investment firm: strategy, portfolio, execution, risk, compliance, debate and accounting agents beneath an AI chief investment officer, communicating only over a bus. Every change to the traded mix had to clear criteria fixed before evaluation on point-in-time data, a rule that deleted the project's founding result and a sleeve holding 40% of the weight. What survived compounds at 15.8% over 2018–2024 and 13.0% on an untouched 2008–2017 holdout, against 13.6% and 8.5% for the S&P 500, with a worst drawdown of −19.1% where the index lost −51.9%. A pre-registered A/B trial then turned the same instrument on the component the project believed in most: the LLM debate gate lowered Sharpe from 0.89 to 0.81 and lost to a rate-matched random veto scoring 0.93. Its benefit was a fivefold cut in turnover, not judgment.

Index Terms—Multi-agent systems, large language models, algorithmic trading, momentum, risk management, backtesting methodology, survivorship bias

I. Introduction

Two weaknesses recur in the applied literature on automated trading. The first is structural: a published system typically compresses the whole investment process into one learned mapping from market state to orders [1]. Firms are not built that way. Idea generation, sizing, limit enforcement, compliance, execution and bookkeeping belong to separate desks with mutual veto rights, and that fragmentation is how an institution stops one desk's blind spot from becoming a market position.

The second is evidential. Applied backtests are widely contaminated by survivorship, by lookahead in the universe definition, and by selecting the strategy after seeing how it scored [13], [15]. Tuning a mix until its Sharpe looks respectable is fitting to the test set under another name, and language-model signals make it worse, because the outcomes a backtest tries to forecast may already sit in the model's training corpus [5].

Horizon Capital responds to both. It is a complete runnable system in which the departments of a fund are autonomous agents on a publish/subscribe bus, with three independent gates between a strategy signal and a broker order. Its signal layer is intentionally legible, built from momentum, trend, quality and regime primitives taken from empirical asset pricing [7], [8], [9]; its judgment layer is a panel of language models arguing adversarially [4].

This paper contributes three things. The first is a validation protocol for autonomous trading agents (point-in-time universes, leave-one-out attribution, pre-registered holdout thresholds, deflation for variants tried), reported together with everything it threw away. The second is a pre-registered measurement of whether an LLM debate gate improves realized outcomes, with two controls this literature normally lacks: prompts blinded to the security's identity and date, and a veto rejecting at the measured rate but selecting at random; the gate loses to the second, and that control is the point. The third is an organizational-chart architecture with three nested gates and per-asset-class routing, plus a reconciliation guard recovered from live deployment.

Related work. Reinforcement-learning agents infer positions end-to-end from price features [1]; Horizon Capital inverts that emphasis, taking allocations from rules attributable sleeve by sleeve and confining the learned component to a veto, where failure costs a trade that never happens rather than an unbounded position. The nearest neighbours are firm-shaped agent collections: TradingAgents gives analyst, researcher and risk personas an argument before committing capital [2], FinMem equips one agent with layered memory [3], and debate improves factuality in general-purpose work [4]. This system borrows the debate, then tests the panel against a rate-matched random control instead of assuming its value, and treats sampling error in a Sharpe drawn from one

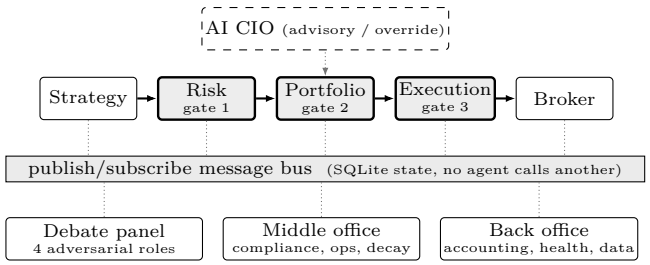


Fig. 1. The firm as an org chart. Solid arrows are the only route from signal to broker order, each a bus message the receiving stage may terminate; the three shaded stages are the nested gates. Debate hangs off the bus: the PortfolioAgent consults it rather than passing through it, so disabling it leaves the path untouched.

path [16] and delisting bias [19] as first-order, because both inflate exactly the statistic such systems are judged on.

II. System Architecture

Each department of a fund is an independent asynchronous agent (Fig. 1). The front office holds the StrategyAgent (signals), the PortfolioAgent (sizing, sole ownership of NAV, custody of the debate gate) and the ExecutionAgent (routing and final guards). The middle office holds the RiskAgent (pre-trade limits plus a 30-second VaR/CVaR/drawdown loop with a sticky kill switch) alongside compliance, operations, the DebateAgent and an alpha-decay monitor. An executive AI chief investment officer issues advisories and may override only while the kill switch is set, and only with high stated confidence.

No agent invokes another. Everything travels as a message on a namespaced publish/subscribe bus with wildcard subscriptions; in local mode that bus is an in-process queue and all state lands in SQLite, so the entire firm runs on one laptop with no infrastructure. That decoupling is what makes the gates enforceable rather than advisory: no edge skips a stage, so every intent passes messages the risk and debate agents can intercept. Risk applies symbol-agnostic weight caps (20% of NAV) and daily-loss limits; the PortfolioAgent sizes against database-backed cash, never the key-value mirror; execution checks kill-switch state, cash sufficiency and a naked-short prohibition, then routes by symbol so no order reaches a venue that does not trade its asset class.

Three modes trade autonomy against oversight: paper trading executes immediately in simulation; human approval queues orders for an interactive decision; fully autonomous makes the debate gate binding and, with no provider reachable, fails closed. The simulator charges commission on every fill and applies an Almgren–Chriss-style temporary-impact term [12], and realized P&L is commission-inclusive, which forces the cash and P&L ledgers to agree exactly.

The firm also survives its own components failing: a supervisor restarts crashed agents from missed heartbeats, exercised in a boot-time database-contention incident that took down four agents and recovered all of them without operator action, and a sticky kill switch halts trading on a drawdown breach until a human clears it. 467 automated tests include regression pins on the mix weights and the cash/P&L identity.

III. Strategy Layer

Signals originate in independent sleeves, each a pure function from historical bars to target weights, blended at fixed proportions; the validated mix holds three.

The research stack (weight 0.500, weekly) gates the book with a Gaussian-mixture regime model on index returns [11]. In risk-off states capital rotates into an inverse-volatility blend of long-duration Treasuries and gold; in risk-on states it ranks the point-in-time S&P 500 by 12–1 time-series momentum [7] and quality [9], holds the top five, and weights them by hierarchical risk parity [10], preferred to mean-variance optimization, whose weights here are dominated by covariance estimation error. Each weight is capped at half-Kelly.

The trend filter (0.333, weekly) applies Faber's 200-day moving-average rule [8] to the index proxy, long above trend and in cash below, with a realized-volatility cap. Momentum rank (0.167, monthly) takes the top three names by cross-sectional 12–1 momentum at equal weight; the live 20% position cap holds its standalone drawdown to -25.4% against -59.5% uncapped. The StrategyAgent emits a signal only when a symbol's net target moves by more than 50 basis points, which is what keeps turnover low. Unvalidated cryptocurrency and prediction-market sleeves are carved out pro-rata only when explicitly enabled, leaving the validated equity mix byte-identical while they are off.

IV. The LLM Debate Gate

Under full autonomy every sized trade must survive a structured argument among four adversarial roles: strategy (the bull case), risk (the bear case), sentiment (news and crowd positioning) and portfolio (fit against existing exposure). Every debater receives the same grounded context and reasons only from it, returning free-text argumentation plus a discrete vote in $\{\text{for}, \text{against}, \text{abstain}\}$ and a confidence in $[0, 1]$. Prose is logged for audit but never parsed for intent, so a debater who argues eloquently and votes against is counted as against. Roles are distributed round-robin across whichever providers are configured, and round two reassigns each debater the opposite role with the first round in context, forcing engagement with the strongest opposing case and reducing same-model agreement bias [4].

Writing F and A for the for and against vote sets across both rounds and c_i for confidences, approval requires both conditions of (1):

$$\begin{aligned} \frac{\sum_{i \in F} c_i}{\sum_{i \in F \cup A} c_i} &> 0.55, \\ \frac{|F|}{|F| + |A|} &\geq 0.6. \end{aligned} \tag{1}$$

The two lines guard different failures: the confidence-weighted share lets a strongly held minority block a lukewarm majority, while the head-count share prevents one very confident debater from carrying the panel alone. Neither denominator counts abstentions, so abstaining shrinks the panel instead of siding with anyone. Malformed output degrades to abstain, and with no provider reachable the debate rejects: under autonomy the system never trades on an absence of judgment.

V. Validation Protocol

Three rules govern every change to the traded mix.

Point-in-time universes. Ranking today's index members backwards through history silently conditions on survival [19]. This project's earliest backtests made that error, and the edge that motivated the whole effort evaporated once membership was reconstructed as of each date. It is a headline here because the machinery that exposed it is the deliverable.

Leave-one-out attribution. Every sleeve is pulled from the mix in turn and the remainder re-evaluated on point-in-time data. A sleeve keeps its weight only if removing it does damage.

Pre-registered holdout gates. Candidates are chosen on 2018–2024 and must then satisfy criteria written down before evaluation on an unseen 2008–2017 holdout that spans the financial crisis: Sharpe no worse than the incumbent, drawdown no deeper, comparable CAGR. Failures ship disabled with the adverse number recorded. No parameter is ever tuned on holdout data.

VI. Performance

Every figure below was regenerated from the repository's harnesses on yfinance daily bars, with one fill model and one risk-free series across all rows. That series matters: the realized 13-week Treasury bill averaged 0.31% across 2008–2017 and 2.33% across 2018–2024, so a fixed hurdle would punish the calmer strategy rather than be conservative. All ratios use the realized daily bill rate [17].

Comparing only against a long-only index is a weak test, so Table I carries two peers: QAI, which replicates aggregate hedge-fund returns and, being a fund rather than a self-reported index, avoids that literature's backfill and survivorship distortions [20], and AOR, a conventional 60/40 allocation. Both post-date the holdout start, so the advantage on that window is a lower bound.

Both windows tell one story: the mix earns more than the index while carrying materially less risk, and the surplus is made in bad markets rather than good ones. Over the selection period it compounds at 15.8% against 13.6% at three-quarters of the index's volatility and roughly half its drawdown, on 78% average gross exposure. On the holdout the gap widens to 13.0% against 8.5%: \$100,000 ends that decade at \$340,322 rather than \$226,948, with a −19.1% worst drawdown where the index lost −51.9%, and 360 trading days underwater instead of 835.

Down-capture localises the edge: on the holdout the mix absorbed 47% of the index's down-day moves, taking under half the damage through the worst equity market since 1929, which is why Sortino (1.35) exceeds Sharpe (0.99) by so much [18]. Against the peers the mix leads on return and on every risk-adjusted measure in both windows; the industry proxy's 25% down-capture is genuinely better, but its Calmar of 0.41 is not, because that protection is paid for in compounding, 3.1% against 13.0%.

A. Attribution and the Sleeve That Was Dropped

Table II decomposes the ensemble. Under the pre-registered rule, leave-one-out is decisive and standalone performance is not, so a sleeve whose removal raises mix Sharpe is a drag regardless of how it looks alone. The breadth-timing sleeve raised it from 0.79 to 0.89. It had carried 40% of the weight. It was removed.

The rest is why an ensemble is warranted. The research stack is load-bearing: taking it out costs 0.17 of Sharpe, more than any other sleeve adds. The trend filter is the interesting case, worst in the book standalone at −0.05 because it sits in cash most of its life; removing it costs only 0.03, but its solo drawdown of −4.0% against the ensemble's −18.0% is what it is bought for. Sleeve correlations span 0.40–0.63, which is the diversification being purchased. A top-10 variant of momentum rank was tested under the same rule, failed, and was discarded in favour of top-3.

B. Holdout Gate and Multiple Testing

The holdout rows double as an acceptance test. Two thresholds were fixed before the candidate saw 2008–2017: G1, Sharpe strictly above the incumbent's; G2, drawdown no more than five points deeper. Both passed with room, 0.99 against 0.88 and a drawdown 7.8 points shallower, and the candidate reached the incumbent's 13.0% CAGR on lower volatility (12.9% against 14.8%) and lower gross exposure (75% against 94%).

A Sharpe ratio chosen as the best of several attempts is biased upward, so the estimates are deflated. The probabilistic Sharpe ratio gives the chance that the true Sharpe clears a benchmark after correcting for skew and excess kurtosis [14]; the deflated Sharpe ratio additionally discounts for picking the best of N trials [13], [15]. Rather than assume a flattering trial count, deflation uses the twelve Sharpe ratios this project actually generated, spread −0.72 to 0.77 with variance 0.159; a wide spread deflates harder than a generic assumption would.

The edge survives on both windows, more comfortably on the one that counts. On the selection window (1,760 days, SR 0.89) the probability of a positive true Sharpe is 0.990, of beating the index 0.742, and after deflating for the twelve trials 0.719; on the holdout (2,517 days, SR 0.99) the same three figures are 0.999, 0.941 and 0.846.

TABLE I

Performance across both evaluation windows under one harness, one fill model and one realized risk-free series. Calmar is CAGR over maximum drawdown; down-capture is the strategy's mean return on days the index fell, as a percentage of the index's (lower is better). Final value assumes $100,000 at the window start. †QAI and AOR post-date the holdout start (88% and 91% coverage).

Variant	CAGR	Vol.	Sharpe	Sortino	Calmar	MaxDD	Down-cap.	Final
Selection period, PIT 2018–2024								
Production mix	15.8%	15.3%	0.89	1.16	0.88	**−18.0**%	63%	$278,355
Prior mix (w/ breadth)	14.6%	16.0%	0.79	1.03	0.60	−24.2%	72%	$259,383
SPY buy-and-hold	13.6%	19.5%	0.64	0.78	0.40	−33.7%	n/a	$244,206
Hedge-fund industry (QAI)	2.5%	6.4%	0.05	0.07	0.16	−14.9%	28%	$118,499
60/40 balanced (AOR)	5.9%	11.5%	0.36	0.45	0.26	−22.9%	58%	$149,254
Holdout period, PIT 2008–2017 (unseen at selection)								
Candidate (production)	13.0%	12.9%	0.99	1.35	0.68	**−19.1**%	47%	$340,322
Incumbent (w/ breadth)	13.0%	14.8%	0.88	1.18	0.48	−26.9%	67%	$337,758
SPY buy-and-hold	8.5%	20.3%	0.49	0.60	0.16	−51.9%	n/a	$226,948
Hedge-fund industry (QAI)†	3.1%	6.0%	0.51	0.63	0.41	−7.7%	25%	$130,768
60/40 balanced (AOR)†	10.2%	13.6%	0.77	0.90	0.42	−24.4%	61%	$242,028

TABLE II

Per-sleeve attribution, PIT 2018–2024. LOO is the mix re-run without that sleeve, weights renormalized. Full-mix Sharpe with breadth = 0.79. All four verdicts are unchanged from the flat-rate computation under which the drop decision was taken.

	Standalone		Leave-one-out	
Sleeve	Sharpe	MaxDD	Sharpe	Verdict
Breadth-timing	0.44	−35.2%	0.89	drags (dropped)
Research stack	0.72	−16.4%	0.63	carries
Trend filter	−0.05	−4.0%	0.76	neutral/hedge
Momentum rank	0.70	−25.4%	0.74	carries

The inference stops there: 0.742 is evidence, not proof, and is exactly the number an undeflated Sharpe would have concealed.

C. An Adverse Result and a Live Failure

Prediction-market prices enter the system as first-class symbols with real execution support, on the hypothesis that event probabilities drift as informed flow arrives ahead of resolution. A 7-day probability-momentum sleeve built to harvest that drift, replayed over CLOB history for the highest-volume markets, lost money: −8.1% aggregate return, Sharpe −0.18, drawdown −22.8%, hit rate 43.3%. It ships disabled. A methodology that only ever approves is not a methodology.

The pipeline has also run live against the Alpaca paper API, sleeve signals becoming real orders polled to terminal fills. That launch exposed an underreported failure mode: the local ledger was seeded at $1M while the brokerage account held $100k of paper equity, so sizing against local NAV produced 3.5× accidental margin exposure. The remedy is a startup reconciliation guard comparing broker equity against local NAV on connect, which on a mismatch above 2% alerts and falls back to simulated execution rather than trading a ledger it cannot trust. Any architecture whose internal state can drift from broker truth needs the equivalent.

VII. Isolating the Debate Gate

Everything above measures the rule-based layer, and says nothing about the marginal value of the LLM panel, which is what this section tests.

Design. Two arms consume an identical signal stream under an identical fill model. Arm A disables the gate, so every risk-approved signal executes. Arm B makes it binding, so a signal executes only on an approved conclusion, and a rejected signal leaves that symbol at its previous target, meaning the gate vetoes exits as well as entries. Sizing, limits and execution are shared, so any divergence is the gate's, and sleeve targets are a pure function of bars and date, so both arms consume a byte-identical stream even after their portfolios separate.

Replay, not forward test. The pre-registered metrics need hundreds of trades; live paper trading produces a handful per week. The experiment replays the production mix over 2018–2024 through the real debate engine, and as a validity check fixed in advance arm A had to reproduce the published mix within ±0.02 Sharpe before any model was called; it does, with −18.0% drawdown in both.

Leakage, measured rather than assumed away. A model trained on data after the window may simply recall what a stock did [5]. The primary arm therefore blinds the panel: prompts carry no ticker, no date and no price level, only six scale-free features (momentum percentile, distance from the 200-day average, realized volatility, drawdown from the 52-week high, index correlation, regime label), which describe thousands of (security, date) pairs and leave no retrieval key. A secondary unblinded arm restores ticker and date on the same signals. The live DebateAgent judges only the strongest signals within a bounded daily budget and releases the rest undebated; arm B reproduces that, so the measurement describes the system as deployed.

Arm C, the rate-matched random gate. A veto that rejects a large share of trades mechanically cuts exposure and turnover, and in a trending sample that alone moves Sharpe, so "arm B beat arm A" would establish only that the system traded less, not that the panel chose well. Arm C holds arm B's measured rejection rate while accepting at random; twenty-five such gates form a null distribution arm B must clear to count as judgment.

Acceptance criterion, fixed in advance. The gate stays in

TABLE III
Debate-gate A/B, blinded panel, PIT 2018–2024, identical signal stream and fill model. Arm C holds arm B's rejection rate but chooses at random (25 trials, mean ± s.d.). ‡A sum of independent one-at-a-time effects, not a realizable P&L.

Metric	A (no gate)	B (gate on)	C (random)
Trades executed	1,867	1,510	1,722
Rejection rate	n/a	59.6%	59.6%
Counterfactual P&L‡	n/a	−101%	n/a
CAGR	15.9%	15.9%	n/a
Volatility	15.3%	17.3%	n/a
Sharpe ratio	0.89	0.81	0.93 ± 0.14
Sortino ratio	1.16	1.05	n/a
Max drawdown	**−18.0%**	−25.7%	n/a
Turnover (× NAV/yr)	12.72	2.69	n/a

the autonomous pipeline if arm B's risk-adjusted return is at least arm A's and the counterfactual P&L of rejected trades is non-positive. Otherwise it reverts to advisory-only.

A. Results

The panel concluded 288 debates over 2,304 model calls, approving 118 and rejecting 170: a 59.6% rejection rate at a median 4.6 seconds per decision. Of these, 270 enter arm B, covering 58% of traded weight, strongest signals first, which is how the live cost gate allocates its budget; the rest traded undebated. It was not a rubber stamp: seventy percent of debates split for against against, and the strategy debater, briefed to argue the bull case, voted against four times as often as for (445 to 106) while portfolio and risk sat near balance.

Binding it made the portfolio worse. Table III shows Sharpe falling from 0.89 to 0.81, Sortino from 1.16 to 1.05, and drawdown deepening from −18.0% to −25.7% on identical signals and fills. CAGR was unchanged and the arms finished within $1,200 of each other, so the damage is in the path rather than the destination. The visible change is turnover, collapsing from 12.72× to 2.69× NAV per year: rejecting three signals in five leaves the book holding positions the sleeves had already decided to trim, and gross exposure rose from 78% to 99.8%. The gate did not select trades so much as freeze the portfolio.

The control that settles it. Arm C rejects at the same 59.6% but picks at random. Twenty-five such gates returned a Sharpe of 0.93 ± 0.14, with a 5th–95th percentile band of 0.73–1.19, and the LLM panel beat only 5 of them, an empirical p of 0.80. Fig. 2 shows why this comparison rather than A-versus-B decides the question: the gated curve enters that band immediately, stays inside it for seven years and ends inside it, so at no point does deliberate selection separate from arbitrary rejection at the same rate. Random rejection was in fact better on both mean and median. The gate's contribution here is the contribution of trading less; without this control the turnover reduction would have been written up as a risk-management benefit.

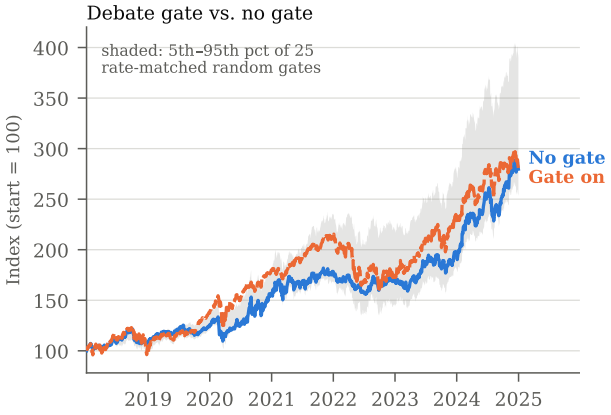


Fig. 2. The debate-gate experiment. Both arms consume an identical signal stream; the gated arm executes only what the panel approved. The shaded band is the 5th–95th percentile of 25 gates rejecting at the same rate but choosing at random, and the gated arm stays inside it throughout.

Was it just remembering? On the 44 signals judged both ways, the unblinded panel approved 34.1% against the blinded panel's 38.6% and reached the same verdict on 72.7% of trades. Revealing ticker and date barely moved anything, which is what the blinded design needs: the panel worked from its features rather than recalling outcomes. The paired sample is small.

The counterfactual. Summed over the 161 rejections with price coverage, the blocked trades would have returned −101% of NAV, and only 38% would have made money. Read naively that says the vetoes were good, which sits awkwardly beside an arm that performed worse. The two do not conflict: the counterfactual adds up each blocked weight change against its own forward return as an independent one-at-a-time event, while the arm those rejections belong to compounds, overlaps and carries the cost of the stale positions the vetoes left standing. Where the two disagree the arm-level result and arm C carry the weight.

Under the criterion of Sec. VII, the gate fails the first condition and reverts to advisory-only. The debate layer keeps its other job: every trade still carries a recorded argument for why it was permitted, which has value in a system trading without a human in the loop. It has not earned authority over execution.

B. What This Says About the Architecture

Two claims are easy to conflate. The gate did not improve returns; the system was indifferent to that, because the gate occupies a veto slot where its failure mode is a trade that does not happen. Arm B's worst outcome across seven years was 0.08 of Sharpe, not a breached limit or an unbounded position. That is the property the three-gate structure exists for: a design in which the least predictable component is load-bearing converts model failure directly into capital loss [6]. Here it is advisory by construction, so the consequence is a deployment posture: the gate ships

enabled in its advisory role, where it generates the audit trail, and disabled as a binding veto until it clears arm C with more than one model family and a sentiment role that has a point-in-time news archive.

VIII. Limitations

No real capital. Everything here is a daily-bar backtest plus a live paper-trading deployment. Peers. QAI and AOR are realized, net-of-fee streams; the strategy rows charge commission and impact but no fees and assume unbounded capacity. Residual survivorship bias. Free daily data omits many delisted securities [19]: 85% of the point-in-time union was recovered on 2018–2024 and 71% on 2008–2017, which leaves absolute levels optimistic, though every arm is affected identically and the ordering holds. One historical path. A Sharpe from a single realization overstates certainty [16], [13]; deflation narrows that without removing it.

The gate experiment. Being a replay, blinding bounds leakage rather than eliminating it: an extreme feature combination can still narrow the era. The panel ran on a single provider, so cross-family diversity is implemented but not demonstrated; the sentiment role abstains structurally for want of a point-in-time news archive, so one voice in four contributes little; and gating the full stream, rather than the deployed budgeted subset, is unmeasured. Training-corpus priors. LLM judgments inherit biases that cannot be audited for a closed model; the blinded/unblinded agreement rules out one specific failure, not whether those priors suit the period judged.

IX. Conclusion

Horizon Capital runs. A firm of decoupled agents with three nested gates between a signal and a broker order, operating on a laptop against a real brokerage API, produces a book that would have turned $100,000 into $340,322 across the financial crisis while the index reached $226,948. The part worth defending is not the return but the protocol that decided what was allowed to stay in the system, because that protocol removed more than it added: the edge that started the project, once point-in-time membership revealed it as survivorship bias; a breadth sleeve holding 40% of the mix; a top-10 momentum variant; and a prediction-market sleeve. What survived returns Sharpe 0.89 and 0.99 against the index's 0.64 and 0.49 at roughly half the drawdown, with deflated probabilities of positive true Sharpe of 0.719 and 0.846.

The same protocol then removed the component the project had been most confident about: under blinded prompts and a control rejecting at the same rate but choosing at random, the debate gate failed to beat the control, and the improvement that would otherwise have been claimed was a fivefold turnover reduction wearing a different hat. That control belongs in any study of an LLM gate, filter or veto, because such mechanisms move performance by themselves. None of this shows that language models cannot contribute to a trading system; it shows that this gate, on these signals, did not, and that the only reason anyone knows is that it was checked. Next steps: multiple model families, a point-in-time news archive for the sentiment role, and the full signal stream.

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